

Lab 9: Thermocouple Calibration

- Review
 - **Safety:**
 - * Keep water away from electronics and voltage sources
 - * Be careful to avoid spills
 - * How water can burn.
 - Thermocouples
 - * Thermocouple fundamentals
 - Thermocouple - “If any two metals are connected together ... a voltage is developed that is a function of the temperature of the junction. This junction of two metals used as a temperature sensor is called a thermocouple [3].”
 - * Problem -
 - In the simplest of thermocouple circuits, there are three junctions that can develop a voltage.
 1. thermocouple wire and thermocouple wire
 2. thermocouple wire and low input to voltmeter
 3. thermocouple wire and high input to voltmeter
 - Typically, we are not interested in the temperature at the interface between voltmeter and thermocouple wires
 - A second thermocouple wire junction (reference junction) at constant temperature can counter this problem.
 - * Thermocouple voltages are small and susceptible to noise. Recommendations:
 - “Thermocouple wire pairs should be twisted to reduce noise. Also, a differential-ended connection is preferred between the thermocouple and the DAS board [1].”
 - With a differential connection “... the thermocouple becomes an isolated voltage source ...” and “the measured signal may drift or suddenly jump ...” “Usually, this interference behavior is eliminated by placing a 10-k Ω to 100 k- Ω resistor between the low terminal of the input and the low-level ground.” [1]
 - Ground loops contribute to common mode noise. “... avoid grounding the tip of the thermocouple ...” See (link).
 - Instrumentation amplifiers
 - * Circuit typically composed of 3 operational amplifiers and several precision resistors.

- * Amplifies the difference between two inputs (differential inputs)
- * Instrumentation amplifiers have a very high ‘Common Mode Rejection Ratio,’ that is their Differential Mode Gain is much larger than their Common Mode Gain; see, for example, [2]
- * Differential vs. Common Mode Gain - often amplifier circuits are used to amplify the difference between two input signals. ‘Differential Mode Gain’ is a term used to describe how much an amplifier amplifies the difference between the signals. Though typically not desired, amplifier circuits also amplify signals common to both inputs quantified by the ‘Common Mode Gain.’ See, for example, [2].

- Equipment

- Digital Multimeter Pro DMM (link)
 - * Thermocouple
- National Instruments USB 6341(link)
- Breadboard with triple power supply
- Sweep/function generator (link)
- Oscilloscope (link)
- 741 op amp (link)
 - * $V_S^+ = 6\text{ V}$
 - * $V_S^- = -6\text{ V}$
- OP07 op amp (link)
 - * $V_S^+ = 6\text{ V}$
 - * $V_S^- = -6\text{ V}$
- INA121PA-ND op amp (link)
 - * $V_S^+ = 6\text{ V}$
 - * $V_S^- = -6\text{ V}$
- Resistors:
- Capacitors:
- K type thermocouple wire
- alcohol thermometer
- water
 - * be cautious not to spill water (we are dealing with electricity)
 - * ice water

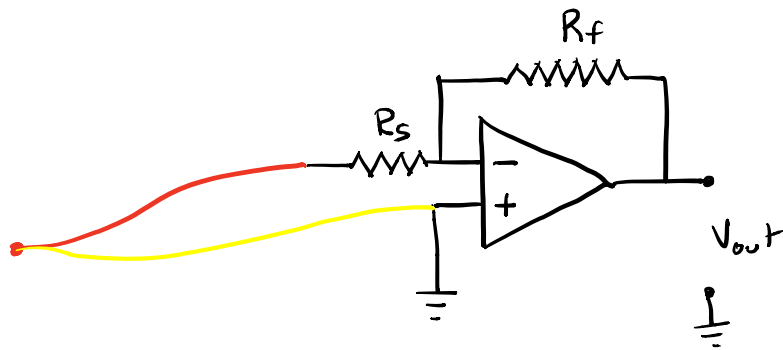
- * hot water - the water and electric kettle are hot and pose a burn risk. Be careful.
- **The objective of this lab is to develop, calibrate, characterize, and demonstrate a thermocouple based system to display on your computer screen in real time water temperatures between 0°C and 100°C with accuracy better than 0.5°C.** To meet this objective, consider the following:
 - Determine if/how you want to condition your signal
 - * Filter (passive, active, digital, order, cutoff frequency)?
 - * Amplification, how much if any (op amp and/or instrumentation amplifier)?
 - * Quantify the amount of noise in your system
 - * How do you limit the effects of noise.
 - DAQ board setup
 - * Sample rate
 - * Range
 - Determine how to arrange your circuit
 - * Number and type of junctions (including reference junctions)
 - * Length of thermocouple wire
 - * Twisted/untwisted wire
 - Determine the sensitivity of your system
 - Determine the response time of your system
 - Determine how to calibrate your sensor
- Lab report:
 - Your lab report should follow the required lab report structure for this class. **However, the body of your report is optional this week.**
 - **The objective of this lab is to develop, calibrate, characterize, and demonstrate a thermocouple based system to display on your computer screen in real time water temperatures between 0°C and 100°C with accuracy better than 0.5°C.**
 - **In the body of your report, you should build a strong case for your objectives. (optional)**
 - Include the following in the Appendix:
 - * This lab outline (required)
 - * Data you collected during lab (required)
 - * Screenshot of each LabView block diagrams (required)

- * Responses to questions in the outline above. (optional)
- * Any other details (data, derivations, calculations, experimental procedure, screen captures etc.)

(The purpose of the Appendix in these reports is to: (i) serve as somewhat of a lab notebook, (ii) demonstrate to the grader what you did during lab, and (iii) hold material that didn't fit into the body of the report but would be necessary to repeat the experiments or justify the claims in the report.)

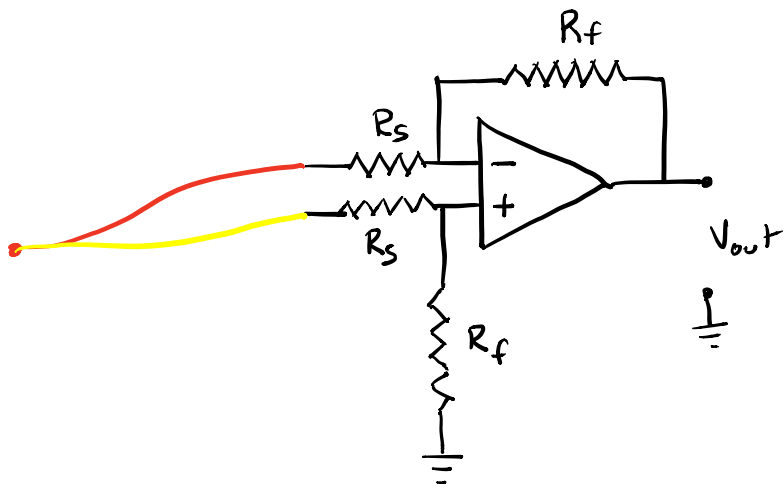
Inverting amplifier

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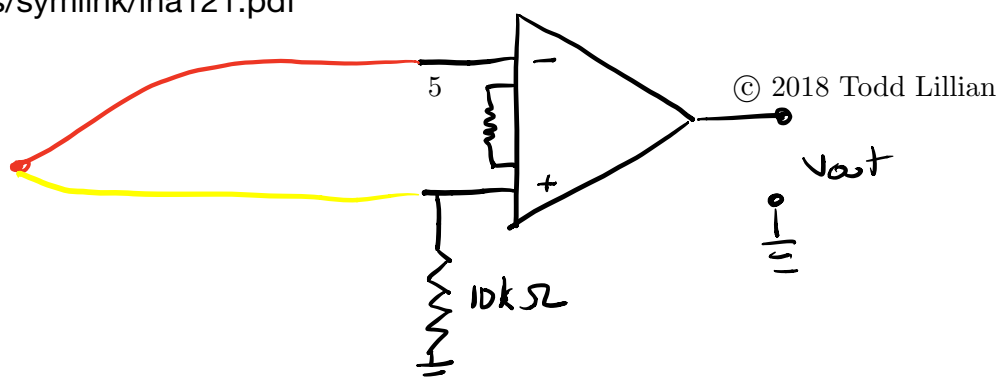
Differential amplifier

www.analog.com/media/en/



Instrumentation amplifier

See: www.ti.com/lit/ds/symlink/ina121.pdf



References

- [1] Richard S. Figliola and Donald E. Beasley. *Theory and Design for Mechanical Measurement, 3rd Ed.* JOHN WILEY & SONS INC, 2000.
- [2] Georgio Rizzoni. *Principles and Applications of Electrical Engineering, 3rd ed.* McGraw-Hill Companies, Inc, 2000.
- [3] Anthony J. Wheeler and Ahmed R. Ganji. *Introduction to Engineering Experimentation, 3rd Ed.* Prentice Hall, 2010.